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dia: An R package for the National Oceanic and Atmospheric Administration dam impact analysis

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Summary

Populations of anadromous (sea-run) fishes such as Atlantic salmon *Salmo salar* have experienced severe global declines as a result of pollution, overfishing, and construction of dams (Limburg & Waldman, 2009). Life-history-based simulation models are commonly used for planning and implementing fisheries recovery activities for many diadromous species (Barber et al., 2018; e.g., Nieland et al., 2013; Stich et al., 2019; Zydlewski et al., 2021). Prior to the expansion of open-source development, many institutionalized decision-support tools utilizing such models relied on closed-source or paid software. For example, the National Oceanic and Atmospheric Administration (NOAA) Dam Impact Analysis (DIA) was originally created as a stochastic life cycle model for Atlantic salmon in the `@RISK` add-in within Microsoft Excel (Nieland et al., 2013). This tool differs from those hosted in similar R packages for non-salmonid species (e.g. Stich et al., 2019; Zydlewski et al., 2021) in that it employs homing and straying probabilities to natal streams for adult spawning and integrates results of physical modeling to inform population dynamics (Nieland et al., 2013; Nieland & Sheehan, 2020). This class of tools, in general, provides advantages for decision making related to anadromous species because it allows integration of geographically and temporally explicit stock dynamics (e.g., influences of dams) that are not readily implemented in classical fisheries stock assessment tools such as those available in existing R packages (Erickson et al., 2022; e.g. Kell et al., 2007; Ogle et al., 2022). We created the `dia` package (Stich et al., 2021) for the R programming language (R Core Team, 2024) as a freely accessible, open-source implementation of these tools that will promote transparency in planning and decision making and improve the availability and quality of these tools over time.

Statement of need

`dia` is an R-based implementation of a previously closed-source life cycle model of Atlantic salmon population dynamics that is used to understand the sensitivity of species recovery trajectories to natural and anthropogenic marine and freshwater influences. It was developed to assess the sensitivity of restoration outcomes to uncertainty in life-history inputs alongside the impacts of dams and restoration activities. The DIA model uses empirical life-history estimates (e.g., survival), predictive flow and resulting flow-specific dam survival modeling, and other empirical data in freshwater and marine environments to simulate consecutive generations of Atlantic salmon in the Penobscot River (Maine, USA) under varying environmental conditions or management decisions. As the largest remaining population of this critically endangered species in the USA, the population is intensively managed. Management decisions include fish passage rates at dams, fishery harvest rates, and numbers and locations for hatchery stocking of fish (Nieland et al., 2013). Since development, it has been used for mechanistic exploration of key life-history uncertainties within the context of species recovery (Nieland et al., 2015)

and to support decision-making at federally regulated hydropower dams on the Penobscot River (e.g., National Marine Fisheries Service, 2013; Nieland & Sheehan, 2020).

We created `dia` for use by fisheries researchers, managers, and practitioners interested in understanding population dynamics of intensively managed endangered Atlantic salmon in the USA. The R package maintains the core routines from the original closed-source version of the model by replicating spreadsheet-based calculations, and incorporates original data and parameter sets as built-in objects that serve as default values for arguments of the primary user-facing functions. However, it also allows exploration of future restoration scenarios through a variety of user-facing options and allows exploration of uncertainty associated with those parameters.

The two primary user-facing functions within the `dia` package are the `run_dia()` and `run_dia_shiny()`, which provide redundant interfaces for using Dam Impact Analysis models in different ways. The `run_dia()` function provides an extensible interface to DIA that can be used for long-run simulation or decision-optimization studies, and allows incorporation of user-specified data sets such as flow-correlated survival probabilities at dams and in free-flowing river reaches, marine survival and other life-history inputs, or fish-stocking data. The `run_dia_shiny()` function deploys a graphical user interface using the `shiny` package (Chang et al. (2022)) that is less extensible but more easily used by fishery managers and practitioners who may be less familiar with programming. `run_dia_shiny()` also includes exportable results from simulation models including `.csv` or other flat-file formats and default plots through the `ggplot2` R package [Wickham (2016); Wickham et al. (2019); Figure 1]. Both can be deployed on networked servers as any other R or `shiny` application to improve accessibility or facilitate use on high performance computers for large simulations. The GitHub repository (Stich et al., 2021) includes additional instructions for installation and a variety of potential uses of `run_dia()` and `run_dia_shiny()`, with shorter examples included in the package help files. While implementation is currently limited to the Penobscot River as a priority conservation water in the USA, the package serves as one example to help generalize these modeling approaches to Atlantic salmon and other sea-run fish in watersheds globally. Specifically, while many of the built-in datasets and helper functions in `dia` are generalized or could be used to simulate life-history information for other systems and species, generalizing the geographic component (i.e., structural river system) represents an important priority for future development.

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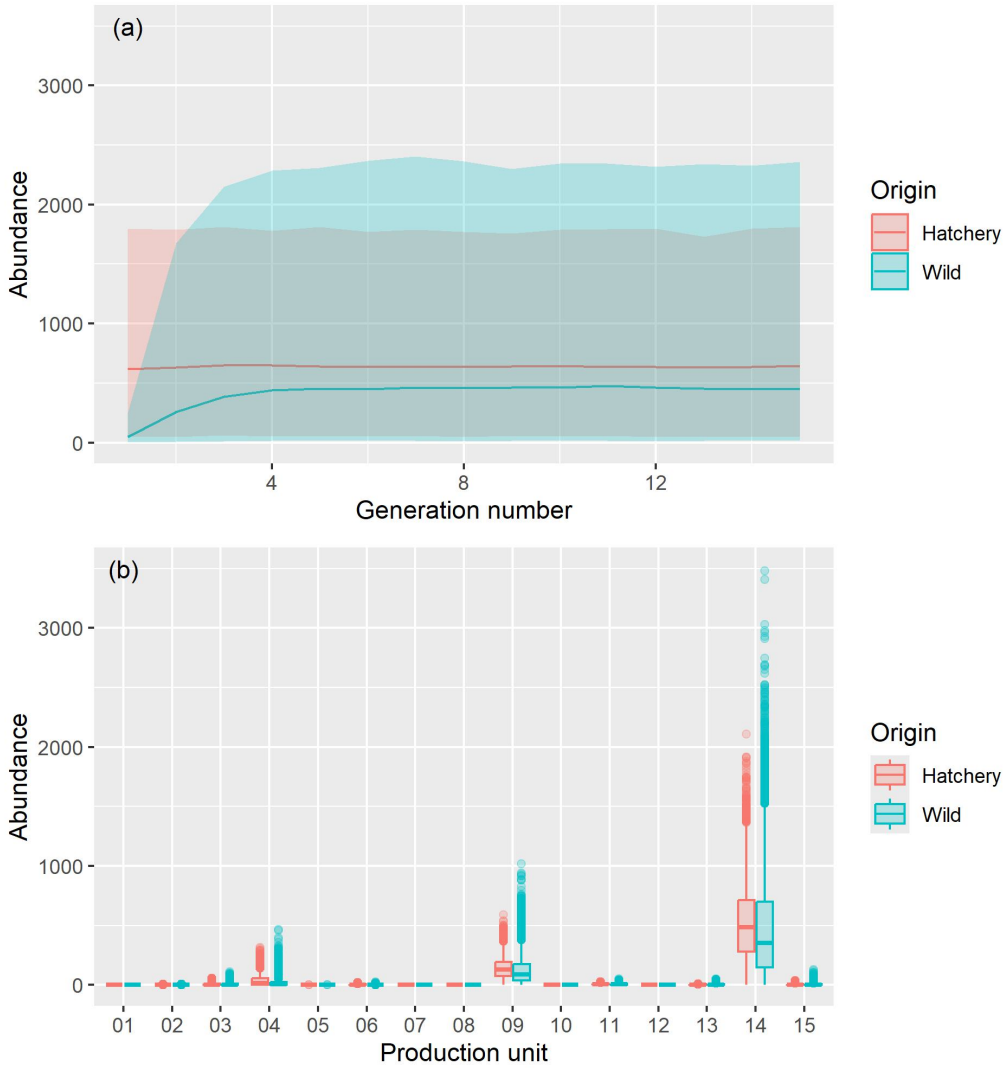

Figure 1: Example graphical outputs using the default argument values (Nieland & Sheehan, 2020) for `run_dia()` to run 10,000 simulations, showing (a) the number of two-sea-winter adult females of hatchery or wild origin returning to the watershed during each generation to spawn, and (b) the number of those fish returning to each production unit within the watershed after 15 generations.

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